HU Collection

TMUA Mock Set D Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	B	D	E	B	E	A	F	A	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	D	C	E	D	B	A	D	B	E	F

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer C.

Write each as a logarithm and pull the powers out. $a = \log_4 9 = \log_2 3$, $b = \log_{27} 25 = \frac{2}{3}\log_3 5$ and $c = \log_{125} 64 = \frac{6}{3}\log_5 2 = 2\log_5 2$. So

$$abc = \tfrac{4}{3}\,\log_2 3 \cdot \log_3 5 \cdot \log_5 2 = \tfrac{4}{3},$$

because the chain $\log_2 3 \cdot \log_3 5 \cdot \log_5 2$ is $\frac{\ln 3}{\ln 2} \cdot \frac{\ln 5}{\ln 3} \cdot \frac{\ln 2}{\ln 5} = 1$. The only work is keeping track of the factors $\frac{2}{3}$ and 2.

Question 2. Answer B.

Stationary points of $y = 3-2f(4-2x)$ occur where $4-2x$ is 2 or -1, that is at $x = 1$ and $x = \frac{5}{2}$. The inner map $x \mapsto 4 - 2x$ reverses direction, but a reflection in a vertical line never turns a maximum into a minimum; the factor -2 outside does. So the minimum of f becomes a maximum: at $x = 1$, $y = 3 - 2(-3) = 9$. The maximum of f becomes the minimum $\left(\frac{5}{2}, -5\right)$. The answer is $(1, 9)$.

Question 3. Answer D.

Multiply the brackets before expanding: $(2 + x)\left(1 - \frac{x}{2}\right) = 2 - x + x - \frac{x^2}{2} = 2 - \frac{x^2}{2}$. So the product is $\left(2 - \frac{x^2}{2}\right)^6$, which contains only even powers. The x^4 term comes from choosing $-\frac{x^2}{2}$ twice:

$$\binom{6}{2} 2^4 \left(-\tfrac{1}{2}\right)^2 = 15 \cdot 16 \cdot \tfrac{1}{4} = 60.$$

Expanding the two sixth powers separately and convolving the coefficients also works, but takes five products and invites sign errors.

Question 4. Answer E.

The geometric condition is $(a + 3d)^2 = a(a + 12d)$, which gives $9d^2 = 6ad$, so $a = \frac{3}{2}d$ since $d \neq 0$. Then $S_{20} = 10(2a + 19d) = 220d = 440$, so $d = 2$ and $a = 3$. The terms are 3, 9, 27: ratio 3. The first five terms sum to $3 + 9 + 27 + 81 + 243 = 363$. Notice that the ratio is 3 whatever d is; the sum of 440 is needed only to fix the scale.

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Question 5. Answer B.

The coefficients sum to zero, so $x = 1$ is always a root, and $x^3 - (a+1)x^2 + (a+1)x - 1 = (x-1)(x^2 - ax + 1)$. A repeated root arises in two ways: the quadratic has a repeated root, which needs $a^2 = 4$, so $a = \pm 2$; or the quadratic has 1 as a root, which needs $2 - a = 0$, so $a = 2$ again. The two routes overlap, leaving exactly two values, $a = 2$ and $a = -2$. Counting three is the trap.

Question 6. Answer E.

Write $\tan x = \frac{\sin x}{\cos x}$ (so $\cos x \neq 0$) and factorise rather than divide: $\sin x \left(\frac{1}{\cos x} - 2\right) = 0$. Either $\sin x = 0$, giving $x = -360°, -180°, 0°, 180°, 360°$, or $\cos x = \frac{1}{2}$, giving $x = \pm 60°, \pm 300°$. That is $5 + 4 = 9$ solutions, none with $\cos x = 0$. Dividing both sides by $\sin x$ at the start loses five of them.

Question 7. Answer A.

We need $x > 0$ and $x \neq 1$. Put $t = \log_2 x$, so $\log_x 16 = \frac{4}{t}$ and the inequality is $t + \frac{4}{t} < 5$. Do not multiply through by t without its sign. If $t > 0$, it becomes $t^2 - 5t + 4 < 0$, so $1 < t < 4$, that is $2 < x < 16$. If $t < 0$, multiplying reverses it: $t^2 - 5t + 4 > 0$, which holds for every negative t, so every $0 < x < 1$ works. The solution set is $0 < x < 1$ or $2 < x < 16$.

Question 8. Answer F.

Move everything to one side instead of multiplying by $x - 1$, whose sign is unknown:

$$\frac{2x}{x-1} - x = \frac{3x - x^2}{x-1} = \frac{x(3-x)}{x-1} \geq 0.$$

The critical values are 0, 1 and 3. The expression is positive for $x < 0$, negative on $(0, 1)$, positive on $(1, 3)$ and negative for $x > 3$; it is zero at $x = 0$ and $x = 3$ and undefined at $x = 1$. So the solution set is $x \leq 0$ or $1 < x \leq 3$.

Question 9. Answer A.

The $120°$ angle is the largest angle, so it faces the longest side $x + 2$. The cosine rule gives

$$(x+2)^2 = x^2 + (x+1)^2 - 2x(x+1)\cos 120° = 3x^2 + 3x + 1,$$

so $2x^2 - x - 3 = 0$ and $x = \frac{3}{2}$. The area is $\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2} \cdot \sin 120° = \frac{15\sqrt{3}}{16}$.

Question 10. Answer C.

With $u = 3^x > 0$ the equation is $u^2 - (k+1)u + k = (u-1)(u-k) = 0$. The root $u = 1$ always gives $x = 0$. The root $u = k$ gives a second value of x exactly when $k > 0$ and $k \neq 1$. So there is exactly one solution when $k \leq 0$ or $k = 1$. Excluding $k = 0$ is the usual slip: then $u = 0$ is simply not available, and $x = 0$ is still the only solution.

Question 11. Answer D.

The intersections satisfy $x\left(x^2 - 3x + 2 - m\right) = 0$. The origin is always one; we need the quadratic to have two distinct roots, neither of them 0. Distinct roots need $9 - 4(2 - m) > 0$, that is $m > -\frac{1}{4}$. A root at 0 happens when $2 - m = 0$: for $m = 2$ the quadratic is $x^2 - 3x$, whose root $x = 0$ duplicates the origin, leaving only two points. So the answer is $m > -\frac{1}{4}$ with $m \neq 2$. At $m = 0$ the roots are 1 and 2, which is fine.

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Question 12. Answer C.

The radius is 5. A chord of length 8 lies at distance 3 from the centre, so we need lines through O at distance 3 from $(3,4)$. The line $x=0$ is at distance 3: it meets the circle at $(0,0)$ and $(0,8)$. For $y=kx$ the distance is $\frac{|3k-4|}{\sqrt{1+k^2}}=3$, giving $24k=7$, so $k=\frac{7}{24}$. That is two lines. Writing every line through O as $y=kx$ silently discards the vertical one and gives the tempting answer 1.

Question 13. Answer E.

The curve crosses the axis at 0 and 2 inside the interval, so split there. With $F(x)=\frac{x^4}{4}-2x^2$: $\int_{-2}^{0}=F(0)-F(-2)=4$, $\int_{0}^{2}=F(2)-F(0)=-4$, and $\int_{2}^{3}=F(3)-F(2)=\frac{9}{4}+4=\frac{25}{4}$. The total area is $4+4+\frac{25}{4}=\frac{57}{4}$. Integrating straight across from -2 to 3 gives $\frac{25}{4}$, because the first two pieces cancel.

Question 14. Answer D.

Let $AC=b$. The cosine rule at A gives $36=100+b^2-20b\cdot\frac{\sqrt{3}}{2}$, that is $b^2-10\sqrt{3}\,b+64=0$. Both roots are positive, and they differ by $\sqrt{300-256}=2\sqrt{11}$. The area is $\frac{1}{2}\cdot 10\cdot b\cdot\sin 30^\circ=\frac{5}{2}b$, so the areas differ by $\frac{5}{2}\cdot 2\sqrt{11}=5\sqrt{11}$. Treating the cosine rule as a quadratic in the unknown side handles the ambiguous case with no need to find either angle.

Question 15. Answer B.

The tangent at $x=t$ is $y=(3t^2-3)(x-t)+t^3-3t$. It passes through $(2,2)$ when $2=(3t^2-3)(2-t)+t^3-3t$, which simplifies to $t^3-3t^2+4=0$, that is $(t+1)(t-2)^2=0$. The double root $t=2$ is a single tangent (the point lies on the curve and this is its own tangent), and $t=-1$ gives a second. So there are two tangents. Counting roots with multiplicity gives three.

Question 16. Answer A.

$\frac{dy}{dx}=2x\left(2x^2-6x+k\right)$, so $x=0$ is always stationary and we need the quadratic to add nothing new. If $k>\frac{9}{2}$ it has no real roots: exactly one stationary point. If $k=\frac{9}{2}$ it has the double root $x=\frac{3}{2}$; the gradient is zero there, so it is a second stationary point, a point of inflection. If $k=0$ its roots are 0 and 3, and 3 is new. So the answer is $k>\frac{9}{2}$. Including $\frac{9}{2}$ confuses "stationary point" with "turning point".

Question 17. Answer D.

$M=A+\frac{1}{3}(B-A)=\left(\frac{20}{3},\frac{4}{3}\right)$, so OM is the line $y=\frac{1}{5}x$, while AC is $x+y=6$. They meet at $P=(5,1)$, three quarters of the way from O to M. Triangle OPC has vertices $(0,0)$, $(5,1)$ and $(2,4)$. Enclose it in the rectangle $[0,5]\times[0,4]$ of area 20 and remove the three right triangles around it, of areas $\frac{5}{2}$, $\frac{9}{2}$ and 4: the triangle has area 9. The parallelogram has area $6\times 4=24$, so the fraction is $\frac{9}{24}=\frac{3}{8}$.

Question 18. Answer B.

The whole region has area $\int_0^4(4x-x^2)\,dx=\frac{32}{3}$. The line meets the curve again where $4-x=k$. The part between the curve and the line has area

$$\int_0^{4-k}\left((4-k)x-x^2\right)dx=\frac{(4-k)^3}{6},$$

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the familiar $\frac{w^3}{6}$ for the segment of $y = x(w - x)$. Setting this equal to $\frac{16}{3}$ gives $(4 - k)^3 = 32$, so $4 - k = \sqrt[3]{32} = 2\sqrt[3]{4}$ and $k = 4 - 2\sqrt[3]{4}$. Using a square root instead of a cube root gives option A.

Question 19. Answer E.

Let $f(x) = x^2 - 2kx + k + 6$. Both roots lie in $(-3, 3)$ exactly when the roots are real, the vertex $x = k$ lies in $(-3, 3)$, and $f(-3) > 0$ and $f(3) > 0$. Real roots: $k^2 - k - 6 \geq 0$, so $k \leq -2$ or $k \geq 3$. Vertex: $-3 < k < 3$. $f(3) = 15 - 5k > 0$ gives $k < 3$, and $f(-3) = 7k + 15 > 0$ gives $k > -\frac{15}{7}$. Intersecting, $-\frac{15}{7} < k \leq -2$. At $k = -2$ the repeated root $x = -2$ is inside the interval, so the endpoint is included. The condition $f(-3) > 0$ is the one most often forgotten, and it is the only one that cuts the interval from below.

Question 20. Answer F.

The squared distance from (x, x^2) to $(0, c)$ is $x^2 + (x^2 - c)^2 = u^2 + (1 - 2c)u + c^2$ with $u = x^2 \geq 0$. This quadratic in u has its vertex at $u = c - \frac{1}{2}$. If $c \leq \frac{1}{2}$ the vertex is at $u \leq 0$, so the minimum over $u \geq 0$ is at $u = 0$, the origin. If $c > \frac{1}{2}$ the minimum is at $u = c - \frac{1}{2} > 0$, two points either side of the axis, strictly closer than the origin. So the answer is $c \leq \frac{1}{2}$. Intuition says the origin wins until c is about the radius of curvature, and the radius of curvature of $y = x^2$ at the origin is exactly $\frac{1}{2}$.

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